

Solution Requirements

Date	1 October 2026
Team ID	SWTID-2026-6275
Project Name	Legal Ease: AI-Powered Legal Document Generator
Team Name	Cipher
Team Leader	U. Harikrishnan
Team Members	N. Manikandan, B. Sri Harani Devi, K. Kesavan
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register with email/phone and log in using password plus OTP (optional Google sign-in). Passwords are hashed and sessions use JWT tokens with auto-expiry.	High
2	Authorization levels	Roles: Guest (view templates), Registered User (create, edit, download own documents), Lawyer/Reviewer (review and approve drafts), Admin (manage templates, users and logs). Access is role-based (RBAC).	High

3	External interfaces	Connects to a Large Language Model (LLM) API for drafting, e-signature and payment gateway APIs, email/SMS services, and cloud storage for saved documents.	High
4	Transactions processing	User inputs are validated, sent to the AI engine, merged into the legal template, and the generated document is stored and returned. Premium downloads and subscriptions are processed securely via a payment gateway.	High
5	Reporting	Dashboard showing the user's document history and status; export of documents as PDF/DOCX; admin reports on usage, template popularity and revenue.	Medium
6	Business rules, etc.	Every document follows a verified template and mandatory clauses; the system asks guided questions, flags missing or conflicting terms, keeps version history, and shows a disclaimer that AI output is not a substitute for legal advice.	High
7	Compliance to laws or regulations	Complies with the IT Act 2000 and the Digital Personal Data Protection Act 2023 (India), and with GDPR principles; user consent is recorded and documents can be deleted on request.	High
8	<i>Other</i>	Multi-language support (English, Tamil, Hindi), clause suggestion and plain-language explanation of legal terms, and email notifications.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Pages load quickly and AI drafting returns a complete document without long waits.	Page load < 2 s; document generated in < 10 s for 95% of requests
2	Scalability	The system scales horizontally using containers and a managed database as users and documents grow.	Supports 10,000 concurrent users and 1 million stored documents
3	Security & Data Privacy	All data encrypted in transit and at rest; input sanitization, regular vulnerability scans, and no sensitive data used to train models.	TLS 1.3 in transit, AES-256 at rest; zero critical vulnerabilities in OWASP Top 10 scan
4	Reliability & Availability	The platform remains available with automatic failover and daily backups.	99.9% uptime per month; recovery time < 1 hour
5	Usability & Accessibility	Simple, mobile-friendly interface with guided steps and support for assistive technologies.	WCAG 2.1 AA compliant; a new user finishes a first document in < 10 minutes
6	Other	Code is modular and well documented for easy updates, and the web app runs on all major browsers and devices.	Test coverage > 80%; works on Chrome, Edge, Firefox, Safari and Android/iOS